CASE STUDY 6: University Hostel and Accommodation Management System

A university plans to develop a Hostel and Student Accommodation Management System to replace its existing manual and spreadsheet-based hostel management process. The proposed system should manage information related to students, hostels, rooms, beds, room allocations, hostel fees, attendance, maintenance complaints, visitors, and disciplinary records. It should support hostel allocation, room management, fee tracking, attendance monitoring, complaint management, and generation of analytical reports for university administration. The system must ensure data consistency, integrity, reduced redundancy, accurate room allocation, and efficient retrieval of hostel information. Analyse the given case study and independently design, develop, and evaluate a relational database solution for the proposed system. The entities, attributes, relationships, table names, and relational schema must be identified and designed by the student based on the requirements of the case study.

1. Requirement Analysis and Database Design

Analyse the case study and:

- Identify the functional requirements and business rules.
- Identify appropriate entities and attributes.
- Identify relationships and cardinalities.
- Design a suitable ER diagram.
- Convert the ER model into a relational schema.
- Identify appropriate primary keys and foreign keys.

2. Database Implementation and Data Management

Implement the proposed database using SQL.

The implementation should demonstrate:

- Appropriate database structure creation and modification.
- Suitable integrity constraints.
- Insertion of realistic student, hostel, room, allocation, fee, and complaint data.
- Modification and deletion of records.
- Enforcement of data consistency and referential integrity.
- Appropriate management of room and bed availability.

The database should contain sufficient interrelated data to support meaningful analysis.

3. SQL-Based Data Processing and Analysis

Develop SQL queries to generate meaningful information from the database.

The queries should address scenarios such as:

- Hostel occupancy.
- Room utilization.
- Student attendance.
- Hostel fee collection.
- Outstanding hostel fees.
- Maintenance complaints.
- Hostel-wise student statistics.
- Frequently reported maintenance problems.
- Disciplinary incidents.
- Students requiring administrative attention.

Appropriate techniques for data filtering, transformation, calculation, grouping, and summarization must be demonstrated.

4. Advanced SQL: Joins, Subqueries and Views

Develop advanced SQL solutions based on the requirements of the case study.

The implementation must include:

- Appropriate joins involving multiple relations.
- Subqueries for solving complex hostel-management requirements.
- At least one correlated subquery.
- At least two meaningful views for presenting derived hostel information.

The student must develop queries such as determining:

- Students having outstanding fees above the average outstanding amount.
- Hostels having occupancy above their capacity benchmark.
- Students with repeated maintenance complaints.
- Students with attendance below hostel average.
- Hostels having higher-than-average maintenance expenditure.
- Students with repeated disciplinary incidents.
- Hostel-level performance.
- Other relevant administrative insights.

The choice and purpose of each advanced query must be explained and justified.

5. Student Hostel Support Risk Identification

Create a Student Hostel Support Risk Identification and Decision-Support Module using the database designed by you. The university wants to identify students who may require administrative intervention based on attendance, outstanding hostel fees, maintenance complaints, and disciplinary history.

Develop an appropriate solution that analyses student hostel information in relation to relevant administrative benchmarks and generates a report such as:

Student	Hostel	Attendance %	Outstanding Fee	Complaints	Disciplinary Incidents	Risk Status
Student A	H01	62	₹18,500	4	2	HIGH PRIORITY
Student B	H03	91	₹0	1	0	NORMAL

The criteria for classifying a student as **HIGH PRIORITY**, **NEEDS ATTENTION**, or **NORMAL** must be defined and justified by the student.

The solution must integrate information from multiple relations and demonstrate effective use of advanced SQL and database-driven decision support.